



Non Functional Requirements

1. Dashboard Performance

For 95% of requests under normal load, the analytics dashboard page must fully load (server response + client render) within 2 seconds.

2. Idempotent Assignment & Email Consistency

The system must guarantee idempotency for reviewer assignments: for a given (paper, round, reviewer) combination, at most one active assignment record may exist, and the system must send at most one invitation email per assignment. Repeated clicks/retries/timeouts must not create duplicate assignments or duplicate emails.

3. Password Security

All passwords must be stored using an adaptive one-way hash with a per-user salt. Password reset must use single-use tokens that expire within 15 minutes. After 5 failed login attempts within 10 minutes, the account must be locked for 10 minutes.

4. Privacy & Access Control (Paper Confidentiality)

A user must be able to view a paper's content (files/links/abstract/history) only if they are an author of that paper or an assigned reviewer for that paper. Otherwise the system must deny access with HTTP 403 and record the attempt in the audit log.

5. Email Delivery Timeliness & Retries

For 95% of cases, invitation and reminder emails must be submitted to the Email Service within 30 seconds of the triggering action. If an email submission fails, the system must automatically retry at least 3 times using exponential backoff, and log each failure and retry attempt.